t3-w06 — condition C1

Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

`entry_behaviour`

• Students have been taught how to draw lines.

`instructional_materials`

• Mathematical set.

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

`current_step`

1

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct parallel lines

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — construct perpendicular lines

`objective_text`

construct perpendicular lines

`current_step`

2

`instructional_events`

• present_content

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — bisect a given line segment

`objective_text`

bisect a given line segment

`current_step`

3

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct line segment

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 90 degrees

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 5 — construct angle 60°

`objective_text`

construct angle 60°

`current_step`

5

`instructional_events`

- present_content
- elicit_performance

`student_tasks`

- construct angle 60 degrees

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

How to Construct Parallel Lines

EXPLANATION

To draw a line that is parallel to a given line, you need a straightedge (ruler) and a set square. A set square has a right angle (90°) and a 45° angle. The 90° side will help you keep the new line the same distance from the original line at every point, ensuring they never meet.

Steps

1. Place the set square so that one of its 90° sides lies along the given line.
2. Hold the set square firmly; the other 90° side will be parallel to the given line.
3. Slide the ruler along the free 90° side of the set square.
4. Draw a line with the ruler. This new line will be parallel to the original line because it follows the same direction as the set square's opposite side.

WORKED_EXAMPLE

Worked Example

Given: A line AB is drawn on your paper.

Goal: Construct a line CD that is parallel to AB.

Solution:

1. Place the set square so that its bottom edge (one side of the 90° angle) lies exactly on line AB.
2. The top edge of the set square, which is also a 90° side, will now be parallel to AB.
3. Hold the set square steady and align a ruler with this top edge.
4. Draw a short line segment CD using the ruler. Line CD is parallel to AB because it follows the direction of the set square's opposite side.

Check: If you extend CD and AB, they will never intersect, confirming they are parallel.

1. You have drawn a line PQ on your paper. Using a set square and a ruler, construct a line RS that is parallel to PQ.
answer: Line RS drawn using the set square's opposite 90° side will be parallel to PQ.

2. A line XY is given. Sketch a line UV that is parallel to XY, describing the steps you would take.
answer: Place the set square with one 90° side on XY, align the ruler with the opposite 90° side, and draw UV along the ruler.

Objective 2

How to Construct a Perpendicular Line

> EXPLANATION
> A perpendicular line meets another line at a right angle (90°). To draw a perpendicular line through a given point on a line, you need a compass and a straightedge.
>
> **Steps**
> 1. Place the compass point on the given point on the line and draw a small arc that cuts the line at two points.
> 2. Without changing the compass width, place the compass point on each of the two intersection points and draw two arcs that intersect above (or below) the line.
> 3. Draw a straight line through the given point and the intersection of the two arcs. This new line is perpendicular to the original line.

> WORKED_EXAMPLE
> **Worked Example**
>
> You have a line AB and need a perpendicular line through point C on AB.
>
> 1. Set the compass to a convenient radius and place the point on C. Draw an arc that meets AB at D and E.
> 2. Keeping the same radius, place the compass on D and draw an arc above the line. Then place it on E and draw another arc that meets the first one at point F.
> 3. Use the straightedge to draw line CF. Line CF is perpendicular to AB because it forms a 90° angle at C.

1. Draw a perpendicular line to the given line PQ at point R (use a compass and straightedge).
answer: A line through R that meets PQ at a right angle, constructed by the three-step method described.

2. Given a line XY and a point Z not on the line, construct a line through Z that is perpendicular to XY.
answer: A line through Z that meets XY at 90°, found by first drawing a line from Z to any point on XY, then using the compass-arc method to locate the right-angle point.

Objective 3

How to Bisect a Line Segment

> EXPLANATION
> To cut a line segment into two equal parts you need to find its midpoint. The classic construction uses a compass:
> 1. Place the compass point on one end of the segment and draw an arc above and below the line.
> 2. Without changing the compass width, move the point to the other end and draw another pair of arcs that intersect the first two.
> 3. The two intersection points (one above, one below) are joined with a straight line. This line crosses the original segment at its midpoint, giving you two equal halves.

> WORKED_EXAMPLE
> Example:
>
> Given segment AB, you want to bisect it.
> - Set the compass to a radius larger than half of AB.
> - With the point on A, draw arcs that cut the plane above and below AB.
> - Keeping the same radius, place the compass on B and draw arcs that intersect the first pair. Label the intersection points C (above) and D (below).
> - Draw the line CD. It meets AB at point M.
> - Measure AM and MB; they are equal, confirming that M is the midpoint.

1. Bisect the line segment PQ using the compass method described above.
answer: The constructed line through the two arc intersections meets PQ at its midpoint, dividing it into two equal lengths.

2. You have drawn a segment XY of length 8 cm. After bisecting it, what is the length of each half?
answer: Each half is 4 cm.

3. If the midpoint of a segment is found at point M, and you measure AM = 5 cm, what is the total length of the original segment?
answer: The total length is 10 cm.

Objective 4

How to Construct a Right Angle (90°) with a Compass and Straightedge

> EXPLANATION
> A right angle measures exactly 90 degrees. You can create it using only a ruler and a compass. The key idea is to use the property of a circle: any angle subtended by a diameter is a right angle (Thales' theorem). By drawing a semicircle on a given line and then marking any point on the semicircle, the lines from the ends of the diameter to that point form a 90° angle.

> WORKED_EXAMPLE
> **Worked Example:**
> 1. Draw a straight line AB – this will be one side of your right angle.
> 2. With the compass point on A, draw an arc that cuts the line AB at point C (choose any convenient radius).
> 3. Without changing the compass width, place the point on B and draw another arc that intersects the first arc at point D.
> 4. Draw a line from A through D. This line AD is the second side of the angle.
> 5. The angle ∠CAD is a right angle (90°) because AD is the line through the point on the semicircle whose diameter is AC.
>
> You can check the construction by measuring the angle with a protractor; it should read 90°.

1. Construct a right angle using the steps above on a fresh sheet of paper. Label the points you use.
answer: A correctly constructed 90° angle with labelled points A, B, C, D as described.

2. Given a line segment XY, use a compass to construct a right angle at point X. List the sequence of actions you take.
answer: 1. Draw a semicircle with diameter XY. 2. Mark any point Z on the semicircle. 3. Draw lines XZ and YZ. 4. The angle ∠ZXY is 90°.

3. Why does any angle subtended by a diameter of a circle equal 90°? Explain in one sentence.
answer: Because the triangle formed is a right triangle by Thales' theorem, where the diameter is the hypotenuse.

Objective 5

Constructing a 60° Angle with a Compass and Ruler

> EXPLANATION
> To draw a 60° angle you only need a straightedge and a compass. Follow these steps:
> 1. Draw a straight line AB and mark a point O on it where the angle will be placed.
> 2. Place the compass point at O and draw an arc that cuts the line at point C.
> 3. Without changing the compass width, move the point to C and draw a second arc that cuts the first arc at point D.
> 4. Draw a straight line from O through D. The angle ∠CO D is 60° because the arcs create an equilateral triangle O C D, and each interior angle of an equilateral triangle is 60°.
> 5. The required angle is the angle between the original line AB and the new line OD.

> WORKED_EXAMPLE
> Example:
> - Draw line AB and choose point O on it.
> - Set the compass to any convenient radius and draw an arc intersecting AB at C.
> - Keeping the same radius, place the compass at C and draw an arc that meets the first arc at D.
> - Join O to D with a straight line.
> - ∠CO D measures 60°.
> You have now constructed a 60° angle at point O.

1. Construct a 60° angle at point P on a given line XY. Describe the steps you would take.
answer: 1. Draw line XY and mark point P on it. 2. With the compass point at P, draw an arc cutting XY at Q. 3. Keep the same radius, place the compass at Q and draw an arc intersecting the first arc at R. 4. Draw a straight line from P through R. The angle between XY and PR is 60°.

2. Using only a ruler and compass, create a 60° angle starting from a given line LM at point M, then label the new ray as MN. What are the construction steps?
answer: 1. Draw line LM and mark point M. 2. With the compass at M, draw an arc intersecting LM at S. 3. Without changing the compass width, place the compass at S and draw an arc that meets the first arc at T. 4. Draw a straight line from M through T; label this ray MN. 5. ∠SMN is 60°, so the angle between LM and MN is 60°.

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	5	Scope	**grounded**
Explanation the principle, rule or relationship to review	5	Method written from event labels — the shape of the teaching without its substance	**substituted**
Worked example a problem with its solution worked through	5	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	12	Set work	**grounded**
Answer what the learner checks against	12	nothing in the input	**unsourced**